TourPlanner –Protocol

Intermediate Submission

Name: Roy Oishik and Krasniqi Erza
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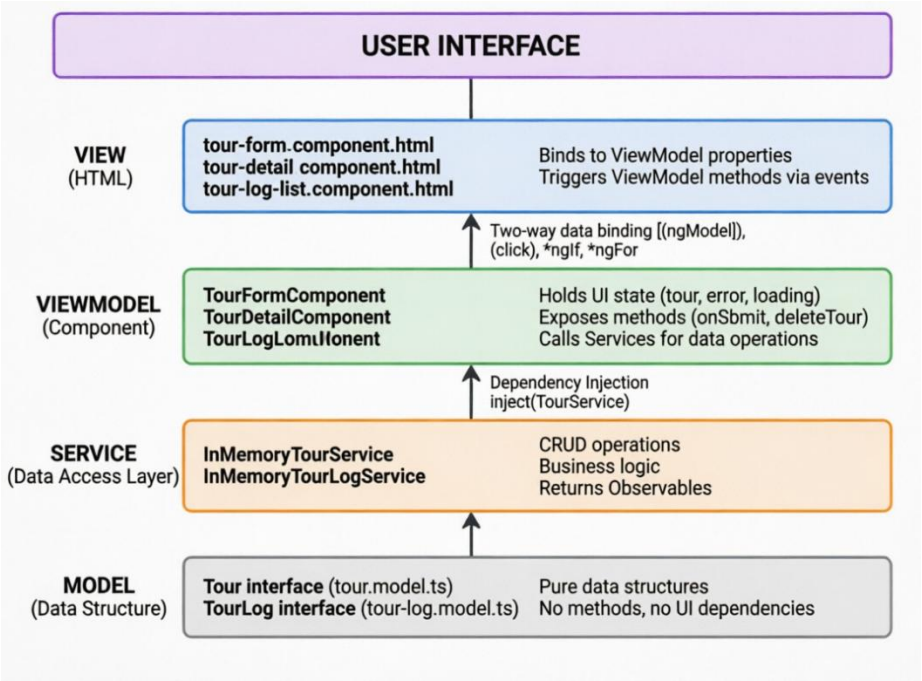
Repository: https://github.com/OishikRoyVienna/Tour-Planner
Project: Tour Planner (Intermediate Hand-In)

Wireframe:

MVVM Pattern Implementation

The application strictly follows the Model-View-ViewModel pattern:



Model Layer (Data Structures)

tour.model.ts

```
export interface Tour {  Show usages  OishikRoyVienna
  id?: number;
  name: string;
  description?: string;
  fromLocation: string;
  toLocation: string;
  transportType: 'BIKE' | 'HIKE' | 'RUNNING' | 'VACATION' | 'CAR';
  distance?: number;
  estimatedTime?: number;
  routeInformation?: string;
  imagePath?: string;
  userId: number;
  popularity?: number;
  childFriendly?: boolean;
}
```

tour-log.model.ts

```
export interface TourLog {  Show usages  OishikRoyVienna
  id?: number;
  dateTime: string;
  comment?: string;
  difficulty?: 'EASY' | 'MEDIUM' | 'HARD';
  totalDistance?: number;
  totalTime?: number;
  rating?: number;
  tourId: number;
}
```

View Layer

View Layer: the view is responsible for displaying data and capturing user input. It contains no business logic, only bindings to ViewModel properties.

Some examples of its use in .html classes:

```html
<!-- tour-form.component.html -->
<div class="form-group">
  <label for="name">Tour Name *</label>
  <input
    type="text"
    id="name"
    [(ngModel)]="tour.name"    <!-- Two-way binding -->
    name="name"
    required
    placeholder="Enter tour name"
  >
</div>
```

```html
<!-- tour-detail.component.html -->
<div class="tour-header">
  <h1>{{ tour.name }}</h1>                    <!-- Interpolation -->
  <img [src]="tour.imagePath" [alt]="tour.name">  <!-- Property binding -->
</div>

<div class="info-item">
  <strong>Distance:</strong> {{ tour.distance }} km
</div>
```

How It Works:

1. **View → ViewModel:** User types "Alpine Hike" → tour.name is updated automatically
2. **ViewModel → View:** Code sets tour.name = "City Tour" → Input field displays "City Tour"
3. **No manual DOM manipulation** required!

ViewModel Layer:

view class tour-form.component.ts at src/app/features/components/tour-form/tour-form.component.ts

that’s a class where made us of components

Showcase of some ViewModel elements at tour-form.component.ts class:

ViewModel Element	Purpose	Bound to View
tour: Partial<Tour>	Holds form data	[(ngModel)]="tour.name"
isEditMode: boolean	Controls UI mode	{{ isEditMode ? 'Edit' : 'Create' }}
error: string	Error message display	*ngIf="error"
saving: boolean	Loading state	[disabled]="saving"
onSubmit()	Form submission handler	(ngSubmit)="onSubmit()"
onCancel()	Cancel action	(click)="onCancel()"